

Avesta MOLAEI

Automotive systems testing engineer.

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Ile de France, Marne la Vallée | Driver's License OK

WORK EXPERIENCE

2025-Present – Test Development Engineer, **Bosch France (VM/ESF4)**, Drancy (93), FRANCE

- *Test Bench Lead for Stellantis activities: wHiL bench setup with IPG CarMaker; full ownership of validation infrastructure, bench design, test development & execution, result analysis*
- *CIP & Lab Development: continuous improvement of test benches/equipment; new electrical standard for bench & lab equipment (NF EN 60204-1, ISO 13849); new lab process for absolute test traceability*
- *General Prototyping: mechanical, electrical & software prototyping of bench/lab equipment & testing tools (e.g. wheel-speed-sensor-in-the-loop rigs, custom CAN/CAN-FD/FlexRay “multiplug” racks, game-controller bench control)*
- *Function Testing: braking SW validation (ISO 26262), spec tracking via IBM ALM/DOORS; Training: apprenticeship & trainee coordination*

2023-2025 – Test Bench Engineer, **Bosch France (VM/EYB5)**, Drancy (93), FRANCE

- *Instrumentation, bench development and management of IIS test campaigns, internal inertial sensor tests on ESP10. Design of a bench testing 2 parts at once, halving test campaign duration*
- *Instrumentation, bench development & management of flowrate campaigns (master cylinder, fluid viscosity)*
- *Development of new testing resources to ensure the continuous improvement of the laboratory*
- *Owned lab's 3D printing division; DASIM simulation & Qlik lab-database referent*

2020-2023 – Prototyping/Virtualization Apprentice, **Bosch France (ETAS/ETFR)**, St Ouen (93), FRANCE

- *ECU/car/environment virtualization; cloud platform (IaaS/SaaS) via Apache Cloudstack, AWS (S3, EC2)*
- *Automotive Software Prototyping (MATLAB/Simulink, eHOOKS, INCA) ; SiL (COSYM/RTPC) ; C & Ruby Programming*

TRAINING

2023-2025 – Bosch France (93)

BR Electrical training (<1000V)

NI DASYlab, Vector Flexray/CAN, EMI trainings

2018-2023 – ESIEE Paris (93)

2018-2023: Engineering diploma (Apprenticeship),

MS in Embedded Systems, Transportation & IoT

- TOEIC (990/990) - Voltaire Certificate

PERSONAL PROJECTS

2023 – **AV-RK MK II** - github.com/avmolaei/AV-RK-MK2

Developed an Open-Source, Ergonomic, Modular Mechanical keyboard

- CAD/CAM, 3D printing, PCB Design and manufacturing, Electronics (SAM21), C programming

2026 – **CANtrip** - avmolaei.com/CANtrip/

Open-source CAN/CAN-FD bus traffic analysis tool

- C++, Reverse engineering, DBC decoding, Wireshark integration, Open-Source

LANGUAGES

- French (*native*)
- English (*fluent*)
- German (*basic*)

TECHNICAL SKILLS

Automotive technologies:

- Braking Systems (ESP10, TMC, IPB/IBooster/BWA)
- Design, installation, and development of test benches
- Mechanics and hydraulics, harnesses and electronics, sensors/actuators/acquisition/measurement.
- Bus CAN, CAN HS, CAN FD, Flexray
- IMC/Famos, CANalyzer, ACW, MM6x, TKwinX, VX/VN, DASIM, TFW, DASYlab, PEAK-PCAN.

Electronics:

Analog and Digital Electronics, Simulation (Cadence, LTSpice), **Design** (Eagle, KiCAD, EasyEDA), **Soldering**
MCU Platforms: PIC16F, ATMEL AVR/Arduino, STM32F4, NXP, ARM Cortex M0, M7

Tools, Machines and CAD/CAM

3D Printing (FDM, SLA), CNC milling, Soldering, Oscilloscopes, Multimeters, Autodesk Fusion 360, CATIA v5

Software skills

C, Java, MATLAB/Simulink, Java for Android
Linux, Network & NAS, Virtualization (VMware, VirtualBox), **Git, GitHub, Cloud.**

HOBBIES

- Electronics: Making, Hacking and DIY
- Automotive: Maintenance and Repairs
- Video Games: History and restoration of antique consoles

References available on request